

DATA SCIENCE PROJECT REPORT

Telco Customer Churn

Exploratory Data Analysis

Churn Drivers · Tenure Analysis · Billing Patterns · Contract & Service Insights

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Executive Summary

This report documents a comprehensive Exploratory Data Analysis (EDA) on the IBM Telco Customer Churn dataset, containing 7,043 customer records from a fictional telecom provider. Churn — the loss of customers to competitors — is one of the most costly business problems in subscription-based industries, making this dataset a high-value analytical target.

This project stands out in the portfolio as the most business-driven analysis. It moves beyond basic descriptive statistics into real retention strategy insights: identifying which service combinations, contract types, and billing patterns are most strongly associated with churn. It also introduces advanced techniques including `pd.to_numeric()` for type coercion, `pd.cut()` for tenure binning, `np.where()` for binary encoding, `pd.get_dummies()` for one-hot encoding, correlation analysis on dummies, KDE density plots, and a custom reusable visualization function.

Project Highlights

Metric	Value
Dataset	IBM Telco Customer Churn (WA_Fn-UseC_-Telco-Customer-Churn.csv)
Size	7,043 rows × 21 columns (raw)

Metric	Value
Post-Cleaning Size	7,032 rows × 19 columns (after dropna + col drop)
Churn Rate	26.5% (1,869 churned out of 7,043)
Class Ratio	73:27 (No Churn : Churn) — imbalanced
Average Monthly Charges	\$64.76
Average Tenure	32.4 months
Max Tenure	72 months
TotalCharges Nulls Found	11 rows (0.15%) — from pd.to_numeric coercion
Tenure Groups Created	6 bins: 1-12, 13-24, 25-36, 37-48, 49-60, 61-72
One-Hot Encoded Columns	pd.get_dummies() applied for correlation analysis
Top Churn Driver	Month-to-month contracts
Top Churn Payment Method	Electronic check
Low Churn Indicator	Long-term contracts (1-2 year), 5+ year tenure
Neutral Factors	Gender, PhoneService availability, MultipleLines
Tools Used	Python, Pandas, NumPy, Seaborn, Matplotlib

1. Introduction

1.1 Business Context

Customer churn — the rate at which customers stop doing business with a company — is a critical KPI in the telecom industry. Acquiring a new customer costs 5–25x more than retaining an existing one. Understanding which customers are at high risk of churning, and why, enables targeted retention strategies such as proactive discount offers, contract incentives, or service upgrades.

This dataset represents a telecom company's customer base with demographic attributes, service subscriptions, contract details, and billing information — all labeled with whether the customer churned in the last month. This makes it ideal for both EDA (what patterns exist in churners?) and future ML modeling (can we predict churn before it happens?).

1.2 Project Objectives

- Profile the dataset structure, types, and completeness
- Handle a real data quality issue: TotalCharges stored as object, not float
- Engineer a new tenure_group feature using pd.cut() for binned analysis
- Perform automated univariate analysis across all categorical predictors using a loop
- Encode the binary target variable Churn using np.where()
- Apply one-hot encoding via pd.get_dummies() for correlation computation
- Identify top churn drivers using correlation bar chart and heatmap

- Analyze billing patterns (Monthly & Total Charges) by churn status using KDE plots
- Perform bivariate analysis: partner, payment method, contract, tech support, and senior citizen vs. churn

1.3 Analysis Questions Answered

#	Question / Analysis Task	Technique Used
1	Shape, columns, dtypes, descriptive stats of raw data	shape, columns.values, dtypes, describe()
2	Churn rate distribution (target variable)	value_counts(), barh plot, percentage calc
3	Null value detection and visualization	isnull().sum(), info(verbose=True), pointplot
4	TotalCharges type coercion (object → float)	pd.to_numeric(errors='coerce')
5	Handle 11 new NaN rows created by coercion	loc filter, dropna(how='any')
6	Engineer tenure_group feature (6 bins)	pd.cut() with custom labels and range
7	Drop customerID and tenure columns	drop(columns=[...])
8	Automated univariate countplot loop (all predictors vs churn)	for loop + sns.countplot(hue='Churn')
9	Encode Churn as binary 0/1	np.where(Churn=='Yes', 1, 0)
10	One-hot encode all categorical variables	pd.get_dummies()
11	Monthly vs Total Charges relationship	sns.lmplot(fit_reg=False)
12	KDE density plots: charges by churn status	sns.kdeplot() with shade=True
13	Correlation of all features with Churn	corr()['Churn'].sort_values().plot(kind='bar')
14	Full correlation heatmap	sns.heatmap(corr(), cmap='Paired')
15	Bivariate: Partner, PaymentMethod, Contract, TechSupport, SeniorCitizen vs Churn	Custom unipLOT() function with hue='gender'

2. Dataset Overview

2.1 Data Source

The IBM Telco Customer Churn dataset is a widely used benchmark in customer analytics and machine learning. It was created by IBM and is available on Kaggle. It represents a fictional telecom company's customer base and is designed to simulate real-world retention analysis challenges.

Repository: github.com/princemaheta2627-glitch/churn_analysis_eda

2.2 Dataset Structure — 21 Raw Columns

Column	Dtype	Category	Description
customerID	object	Identifier	Unique customer ID — dropped before analysis
gender	object	Demographic	Male / Female
SeniorCitizen	int64	Demographic	0 = Not senior, 1 = Senior citizen
Partner	object	Demographic	Has a partner: Yes / No
Dependents	object	Demographic	Has dependents: Yes / No
tenure	int64	Behavioral	Months as customer (0–72) — binned into groups
PhoneService	object	Service	Has phone service: Yes / No
MultipleLines	object	Service	Multiple phone lines: Yes / No / No phone service
InternetService	object	Service	DSL / Fiber optic / No
OnlineSecurity	object	Service	Online security add-on: Yes / No / No internet
OnlineBackup	object	Service	Online backup add-on: Yes / No / No internet
DeviceProtection	object	Service	Device protection: Yes / No / No internet
TechSupport	object	Service	Tech support add-on: Yes / No / No internet
StreamingTV	object	Service	TV streaming: Yes / No / No internet
StreamingMovies	object	Service	Movie streaming: Yes / No / No internet
Contract	object	Contract	Month-to-month / One year / Two year
PaperlessBilling	object	Billing	Paperless billing: Yes / No
PaymentMethod	object	Billing	Electronic check / Mailed check / Bank transfer / Credit card
MonthlyCharges	float64	Financial	Monthly bill amount in USD
TotalCharges	object → float64	Financial	Cumulative charges — stored as object, converted to numeric
Churn	object → int	Target	Whether customer churned: Yes/No → 1/0

2.3 Descriptive Statistics (Numeric Columns)

Statistic	SeniorCitizen	Tenure (months)	MonthlyCharges (USD)
Count	7,043	7,043	7,043
Mean	0.162 (16.2% senior)	32.4 months	\$64.76
Std Dev	0.369	24.6 months	\$30.09
Min	0	0	\$18.25

Statistic	SeniorCitizen	Tenure (months)	MonthlyCharges (USD)
25th Percentile	0	9 months	\$35.50
Median (50th)	0	29 months	\$70.35
75th Percentile	0	55 months	\$89.85
Max	1	72 months	\$118.75

Key observations: 75% of customers have tenure under 55 months; average monthly bill is \$64.76 but 25% pay above \$89.85 — indicating a high-value premium segment. SeniorCitizen's quartile stats are meaningless as it is actually a binary categorical variable (0/1), not a true numeric.

3. Data Cleaning Pipeline

3.1 Step-by-Step Cleaning

Step	Action	Detail	Result
1	Create working copy	<code>telco_data = telco_base_data.copy()</code>	Preserve raw data for reference
2	Convert TotalCharges to numeric	<code>pd.to_numeric(TotalCharges, errors='coerce')</code>	Object → float64; invalid values → NaN
3	Detect 11 new NaN rows	<code>telco_data.loc[TotalCharges.isnull()]</code>	All 11 have tenure=0 (new/trial customers)
4	Drop NaN rows	<code>dropna(how='any', inplace=True)</code>	7,043 → 7,032 rows (0.15% removed)
5	Engineer tenure_group	<code>pd.cut(tenure, range(1,80,12), labels=[...])</code>	6 bins: 1-12, 13-24, ..., 61-72 months
6	Drop customerID & tenure	<code>drop(columns=['customerID', 'tenure'])</code>	Removes ID + raw tenure (replaced by group)
7	Encode Churn to binary	<code>np.where(Churn=='Yes', 1, 0)</code>	Yes → 1, No → 0
8	One-hot encode all categoricals	<code>pd.get_dummies(telco_data)</code>	Creates dummy columns for all object columns

3.2 Key Code: TotalCharges Type Coercion

The most important data engineering step in this project is converting TotalCharges from object to float. This is a common real-world issue where numeric data is stored as strings due to whitespace or formatting:

```

- telco_data.TotalCharges = pd.to_numeric(telco_data.TotalCharges,
    errors='coerce')
- # errors='coerce' turns non-convertible values into NaN
- telco_data.isnull().sum() # TotalCharges: 11 NaN values revealed

```

```
- telco_data.loc[telco_data['TotalCharges'].isnull()] # All have tenure=0
- telco_data.dropna(how='any', inplace=True) # 7043 → 7032 rows
```

3.3 Tenure Binning Feature Engineering

```
- labels = ['{0} - {1}'.format(i, i+11) for i in range(1, 72, 12)]
- # → ['1 - 12', '13 - 24', '25 - 36', '37 - 48', '49 - 60', '61 - 72']
- telco_data['tenure_group'] = pd.cut(telco_data.tenure, range(1,80,12),
    right=False, labels=labels)
```

Tenure Group	Customer Count	Share	Churn Risk
1 – 12 months	2,175	30.9%	Highest — new customers at most risk
61 – 72 months	1,407	20.0%	Lowest — loyal long-term customers
13 – 24 months	1,024	14.5%	Elevated — still in early relationship
25 – 36 months	832	11.8%	Moderate — stabilizing
37 – 48 months	832	11.8%	Moderate — stabilizing
49 – 60 months	762	10.8%	Low — approaching long-term loyalty

4. Key Findings & Insights

4.1 Churn Rate Overview

The target variable Churn is imbalanced at a 73:27 ratio:

Status	Count	Percentage	Implication
No Churn	5,174	73.5%	Majority class — model will favor this without correction
Churned	1,869	26.5%	Minority class — the business-critical group to predict

Business Implication: With 26.5% churn rate, if the company has 1 million customers and each pays \$65/month, that's \$206M/month in recurring revenue at risk from churn. Even reducing churn by 5% through targeted interventions would save approximately \$10M/month. This makes churn prediction one of the highest-ROI machine learning applications in business.

4.2 Monthly Charges — Churn KDE Analysis

The KDE (Kernel Density Estimate) density plot of MonthlyCharges by churn status revealed a critical insight:

- Non-churners: Density peaks at lower monthly charges (~\$20–\$40 range)
- Churners: Density peaks at higher monthly charges (~\$70–\$100 range)
- Clear separation between the two distributions — high charges correlate strongly with churn

Insight: Customers on expensive plans are the most likely to churn — possibly because they feel they are not getting value for their premium spend, or because competitors offer better pricing. This is a direct signal for pricing strategy and retention offers.

4.3 Total Charges — Surprising Insight

The KDE plot for TotalCharges revealed a counterintuitive pattern:

- Churners cluster at LOW total charges (high density at \$0–\$1,000)
- Non-churners have higher total charges (accumulated over longer tenure)

At first glance this seems to contradict the monthly charges finding. However, combining all three variables (tenure + monthly charges + total charges) reveals the full picture:

Factor	Churners Pattern	Explanation
Monthly Charges	HIGH (\$70–\$100)	On expensive plans
Tenure	LOW (1–12 months)	New customers, haven't stayed long
Total Charges	LOW (\$0–\$1,000)	Short tenure × high monthly = low total

Insight: The highest churn risk profile is a new customer (first year) on a high-cost plan. These customers haven't yet built loyalty or seen sufficient value to justify the premium. Early intervention in the first 12 months is the most critical retention window.

4.4 Top Churn Drivers — Correlation Analysis

The correlation bar chart (corr()['Churn'].sort_values()) revealed the strongest positive and negative predictors of churn:

High Churn Drivers (Positive Correlation)

Feature	Direction	Insight
Month-to-month contract	Strong positive	No contractual commitment → easiest to leave
Fiber optic internet	Positive	Premium service with higher bills → value mismatch risk
No online security	Positive	Missing value-add service → less engagement
No tech support	Positive	Unsupported customers feel abandoned → leave

Feature	Direction	Insight
Electronic check payment	Positive	Manual/active payment method → less 'sticky' than auto-pay
First year of subscription (tenure 1-12)	Positive	Highest risk window before loyalty forms
Paperless billing	Positive	Slightly correlated — digital-savvy users may shop around more

Low Churn Indicators (Negative Correlation)

Feature	Direction	Insight
Two-year contract	Strong negative	Long-term commitment = strong retention lock-in
One-year contract	Negative	Medium-term contract still reduces churn significantly
No internet service	Negative	Phone-only customers are simpler, lower-cost, more stable
Tenure 5+ years	Strong negative	Long-tenured customers are highly loyal
Bank transfer / credit card payment	Negative	Auto-pay customers are more passive and less likely to cancel

Neutral Factors (Near-Zero Correlation)

- Gender — Male vs. Female has virtually no impact on churn
- PhoneService availability — whether a customer has phone service does not predict churn
- Number of multiple lines — also near-zero correlation with churn

Business Implication: Gender-based retention campaigns would be wasteful. Resources should focus on contract type upgrades, tech support enrollment, and online security adoption.

4.5 Bivariate Analysis — Churned Customer Profiles

Payment Method (Churned Customers)

Among customers who churned, electronic check was the dominant payment method — significantly higher than mailed check, bank transfer, or credit card. Across both genders, this pattern held consistent.

Insight: Electronic check customers are 'active' payers — they must consciously pay each month, making the cancellation decision more natural. Automatic payment methods (bank transfer, credit card) create inertia that reduces churn.

Contract Type (Churned Customers)

Month-to-month contract customers overwhelmingly dominate the churned population. One-year and two-year contract holders show dramatically lower churn counts.

Insight: This is the single most powerful retention lever. Converting month-to-month customers to annual contracts — even with a small discount incentive — would have outsized impact on reducing churn. Every converted customer represents approximately \$780/year in protected revenue.

Tech Support (Churned Customers)

Among churned customers, the majority had no tech support subscription. Customers without tech support were significantly more likely to churn than those with it.

Insight: Tech support acts as a 'sticky' service — customers who receive support feel more connected and valued. Proactively offering free trial tech support to new customers in the first 6 months could significantly reduce early churn.

Senior Citizen Status (Churned Customers)

Non-senior citizens (SeniorCitizen = 0) make up the majority of churned customers in absolute count. However, given that only 16.2% of customers are seniors, the relative churn rate among seniors is actually higher — seniors churn at a disproportionately higher rate when controlling for base population.

Insight: Senior customers may have more difficulty navigating digital services and are more likely to abandon if they lack support. Dedicated senior customer service initiatives could improve both retention and satisfaction.

Partner Status (Churned vs. Non-Churned Customers)

Among churned customers, those without partners had higher churn rates. Non-churned customers showed a more balanced partner/no-partner split. This suggests customers with partners (and potentially shared plans) have more reasons to stay.

5. Technical Skills Demonstrated

5.1 Advanced Data Engineering

- `pd.to_numeric(col, errors='coerce')` — forcing mixed-type column to numeric, surfacing hidden nulls
- `telco_base_data.copy()` — creating a working copy to preserve raw data integrity
- `pd.cut(col, bins, right=False, labels=[...])` — custom binning of continuous variable into labeled groups
- List comprehension for dynamic label generation: `['{0} - {1}'.format(i, i+11) for i in range(...)]`
- `drop(columns=[...], axis=1)` — removing identifier and redundant columns

5.2 Target Variable Engineering

- `np.where(condition, true_val, false_val)` — vectorized binary encoding without loops
- `pd.get_dummies(df)` — one-hot encoding all categorical columns simultaneously for correlation computation
- Exporting processed DataFrame: `to_csv('tel_churn.csv')` — saving model-ready data

5.3 Advanced Visualizations

- `sns.kdeplot()` with `shade=True`, dual overlay (Churn=0 red, Churn=1 blue) — density comparison
- `corr()['Churn'].sort_values().plot(kind='bar')` — ranked correlation chart with `figsize=(20,8)`
- `sns.heatmap(corr(), cmap='Paired', figsize=(12,12))` — full feature correlation matrix
- `sns.lmplot(fit_reg=False)` — scatter plot without regression line for charge relationship
- `info(verbose=True)` — full column info display for wide DataFrames
- `matplotlib.ticker (mtick)` — imported for advanced tick formatting
- `plt.yscale('log')` — logarithmic y-axis in custom function for better visibility of skewed distributions

5.4 Custom Reusable Function

A custom `uniplot()` function was defined for bivariate analysis, demonstrating software engineering thinking within data science:

```
- def uniplot(df, col, title, hue=None):
-     sns.set_style('whitegrid')
-     sns.set_context('talk')
-     plt.rcParams['axes.labelsize'] = 20
-     fig.set_size_inches(width, 8) # dynamic width based on unique values
-     plt.yscale('log')
-     ax = sns.countplot(data=df, x=col, order=df[col].value_counts().index,
-                        hue=hue)
```

This function is reusable across any DataFrame column, automatically adjusts figure width, applies log scale, and sorts bars by frequency — a professional, DRY (Don't Repeat Yourself) approach to EDA.

5.5 Automated Loop-Based EDA

```
- for i, predictor in
-     enumerate(telco_data.drop(columns=['Churn', 'TotalCharges', 'MonthlyCharges'])):
-     plt.figure(i)
-     sns.countplot(data=telco_data, x=predictor, hue='Churn')
```

This loop automatically generates a countplot for every categorical predictor split by Churn status — producing 17 charts with 3 lines of code. This is the correct professional approach for systematic univariate EDA on multi-column datasets.

6. Conclusion & Retention Strategy Recommendations

6.1 Summary of Findings

- Churn rate is 26.5% — a significant business problem with major revenue implications
- Month-to-month contract is the #1 churn driver — contractual flexibility enables easy exits
- Fiber optic users, despite premium service, churn more — value perception issue
- No online security + no tech support are strong churn signals — disengaged customers
- Electronic check users churn more — active payment creates a natural cancellation window
- High monthly charges + low tenure + low total charges = highest risk customer profile
- Gender, PhoneService, and MultipleLines have no significant impact on churn
- Two-year contracts and 5+ year tenure are the strongest churn protectors
- Senior citizens churn at disproportionately high rates relative to their population share

6.2 Data-Driven Retention Recommendations

Recommendation	Target Segment	Expected Impact	Data Basis
Offer annual contract upgrade incentive	Month-to-month customers	High — strongest churn driver	Contract correlation analysis
Proactive tech support outreach in first 90 days	New customers (tenure 1-12)	High — critical early window	Tenure group + TechSupport bivariate
Auto-pay enrollment campaign with discount	Electronic check users	Medium-High — reduces active cancellation	Payment method bivariate
Online security bundle trial (3 months free)	No-security subscribers	Medium — increases service stickiness	Correlation chart insight
Senior citizen dedicated support line	SeniorCitizen = 1	Medium — high relative churn rate	Senior bivariate analysis
Monthly charge review for fiber optic customers	Fiber optic high-charge segment	High — value mismatch is churn trigger	KDE + correlation

6.3 Potential Next Steps

Enhancement	Description	Technique
Churn Prediction Model	Build binary classifier using cleaned & encoded data	Logistic Regression, Random Forest, XGBoost
Class Imbalance Handling	Address 73:27 imbalance in training	SMOTE, class_weight, stratified k-fold
SHAP / Feature Importance	Explain which features matter most in the model	SHAP values, permutation importance
Customer Lifetime Value (CLV)	Combine churn risk with revenue to prioritize retention	$CLV = ARPU \times (1/churn_rate)$
Survival Analysis	Model time-to-churn as a time-series problem	Kaplan-Meier, Cox Proportional Hazards
Segmentation	Cluster customers into risk profiles for targeted campaigns	K-Means on scaled features

6.4 Learning Outcomes

This was the most advanced and business-relevant project in the portfolio. It introduced production-grade data engineering patterns (pd.to_numeric with coerce, copy() before mutation), sophisticated multi-chart analysis (KDE overlays, correlation ranking, heatmap), and the discipline of building reusable functions instead of copy-pasting code. Most importantly, it demonstrated the ability to translate statistical findings into concrete, actionable business recommendations — the core competency that separates data scientists from data analysts.

7. Project Information

Project Title	Telco Customer Churn Analysis EDA
Author	Prince Maheta
GitHub Repository	github.com/princemaheta2627-glitch/churn_analysis_eda
Notebook File	Churn Analysis - EDA.ipynb
Dataset	IBM Telco Customer Churn (WA_Fn-UseC_-Telco-Customer-Churn.csv) — 7,043 rows, 21 columns
Business Domain	Telecom Customer Retention / Churn Prediction

Platform	Jupyter Notebook / Anaconda (Python 3)
Languages & Tools	Python 3, Pandas, NumPy, Seaborn, Matplotlib
Key Techniques	pd.to_numeric, pd.cut, np.where, pd.get_dummies, KDE plot, corr() heatmap, custom uniplot()
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This report was prepared as part of a data science portfolio.
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